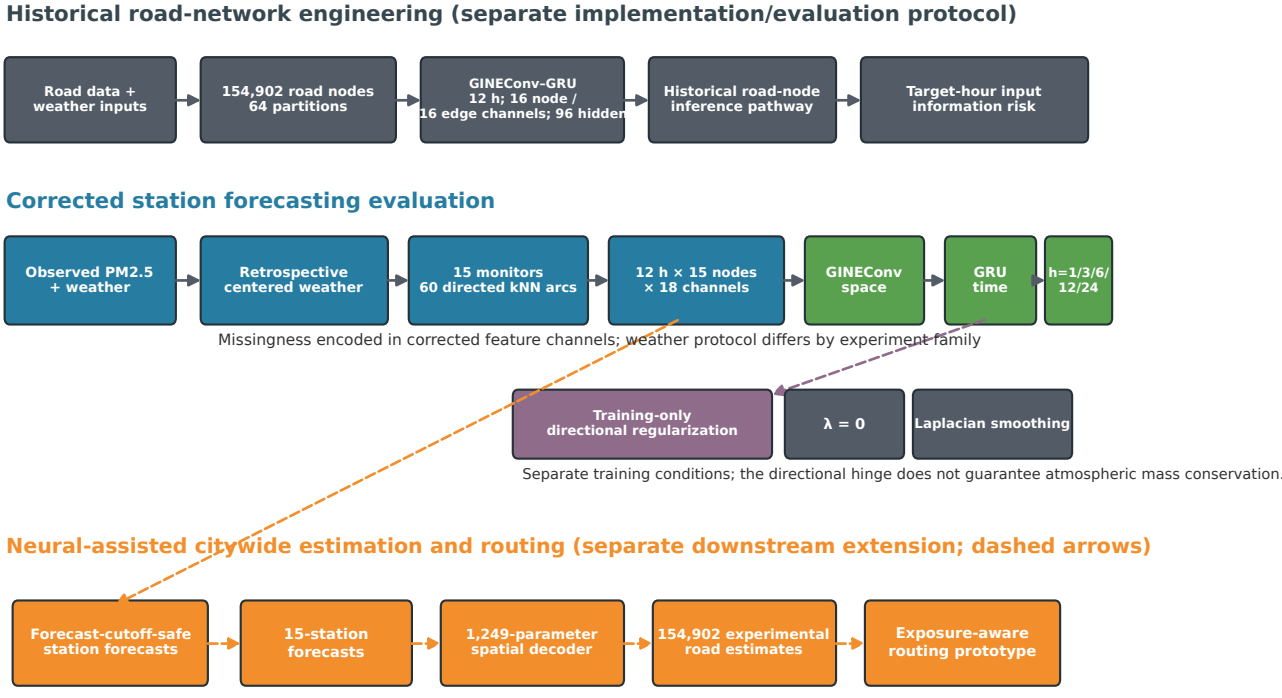
Historical road-network engineering (separate implementation/evaluation protocol)
Road data + weather inputs
154,902 road nodes 64 partitions
GINEConv–GRU 12 h; 16 node / 16 edge channels; 96 hidden
Historical road-node inference pathway
Target-hour input information risk
Corrected station forecasting evaluation
Observed PM2.5 + weather
Retrospective centered weather
15 monitors 60 directed kNN arcs
12 h × 15 nodes × 18 channels
GINEConv space
GRU time
h=1/3/6/12/24
Missingness encoded in corrected feature channels; weather protocol differs by experiment family
Training-only directional regularization
λ = 0
Laplacian smoothing
Separate training conditions; the directional hinge does not guarantee atmospheric mass conservation.
Neural-assisted citywide estimation and routing (separate downstream extension; dashed arrows)
Forecast-cutoff-safe station forecasts
15-station forecasts
1,249-parameter spatial decoder
154,902 experimental road estimates
Exposure-aware routing prototype